



Advanced Time Series Analysis and Forecasting

Seminar 4: Cointegration revisited: VECM, ARDL and panel data



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Acronyms used in this material (1/2)

- ▣ **ADF**: Augmented Dickey–Fuller (test)
- ▣ **AI**: Artificial Intelligence
- ▣ **AIC**: Akaike Information Criterion
- ▣ **ARDL**: AutoRegressive Distributed Lag (model)
- ▣ **BIC**: Bayesian Information Criterion
- ▣ **BNR**: Banca Națională a României (the National Bank of Romania)
- ▣ **BUBOR**: Budapest Interbank Offered Rate
- ▣ **CD**: Cross-section Dependence (test of Pesaran)
- ▣ **CIPS**: Cross-sectionally augmented IPS (test)
- ▣ **ECM**: Error Correction Model
- ▣ **EU**: European Union
- ▣ **EU-27**: the 27 member states of the European Union
- ▣ **FRED**: Federal Reserve Economic Data
- ▣ **GDP**: Gross Domestic Product
- ▣ **HICP**: Harmonised Index of Consumer Prices
- ▣ **HQ**: Hannan–Quinn (information criterion)
- ▣ **IFS**: International Financial Statistics (IMF)

Acronyms used in this material (2/2)

- ▣ **IMF**: International Monetary Fund
- ▣ **IPS**: Im–Pesaran–Shin (panel unit-root test)
- ▣ **KPSW**: King–Plosser–Stock–Watson (common-trends model, 1991)
- ▣ **LLC**: Levin–Lin–Chu (panel unit-root test)
- ▣ **LR**: Likelihood Ratio (test)
- ▣ **MG**: Mean Group (estimator)
- ▣ **NARDL**: Nonlinear AutoRegressive Distributed Lag (model)
- ▣ **PMG**: Pooled Mean Group (estimator)
- ▣ **PSS**: Pesaran–Shin–Smith
- ▣ **RA**: Reinsel–Ahn (small-sample correction)
- ▣ **SE**: Standard Error
- ▣ **TSA**: Time Series Analysis
- ▣ **US**: United States
- ▣ **VAR**: Vector AutoRegression
- ▣ **VAT**: Value Added Tax
- ▣ **VECM**: Vector Error Correction Model

Today's question and route

- **Question:** how do we decide whether non-stationary series share a long-run equilibrium, test what theory says about it, and keep the answer honest in small samples and across countries?
 - ▶ this seminar comes **before** Lecture 4: the section “Prerequisites for Today” gives every definition the tasks use
- **Route**
 - ▶ Part A: the trace test from eigenvalues, degrees of freedom, the Granger representation, a permanent shock, ARDL algebra, a bounds decision, the CD statistic, the Hausman test
 - ▶ Part B: interest-rate pass-through in Hungary; Romanian fuel prices and the oil price; the KPSW common trend on longer samples; EU-27 inflation
 - ▶ Part C: asymmetric fuel-price adjustment (NARDL); an AI answer to audit
- Notebook for today: [open the seminar notebook in Google Colab](#); each task names its notebook section

Exercise map

Task	Question	Type	Model
A1, A2	the trace test from eigenvalues; degrees of freedom of LR tests	Solved, Proposed	A1
A3, A4	the Granger representation; the permanent shock of a common-trends model	Solved, Proposed	A3
A5, A6	an ARDL in error-correction form; a bounds-test decision	Solved, Proposed	A5
A7, A8	the CD statistic; the Hausman test of PMG against MG	Solved, Proposed	A7
B1, B2	Hungarian pass-through: rank tests; tests on β and α	Solved, Proposed	B1
B3, B4	bounds test for fuel prices; the KPSW common trend on longer samples	Solved, Proposed	B3
B5	EU-27 inflation: IPS against CIPS	Proposed	A7, B1
C1, C2	asymmetric fuel-price adjustment; what is wrong in an AI answer?	Proposed	B3, A1–A8

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model

Data used

Series	Source	Frequency	Period
Hungary: lending and deposit rates; BUBOR 3M	IMF IFS; Eurostat	monthly	2005–2026
Romania: HICP fuels; Brent; USD/RON	Eurostat; FRED; BNR	monthly	2007–2026
US consumption, investment, GDP	FRED	quarterly	1949–2025
EU-27 HICP inflation	Eurostat	annual	1999–2025

- In the notebook: `johansen`, `restricted_ml`, `weak_exog_lr`, `boot_trace`, `ardl_ecm`, `bounds_quantiles`, `cd_test`, `ips_tbar`, `cips`; no account or key is needed

Prerequisites for Today (1/8): the VECM

- VECM (TSA, Chapter 7): the change of each variable reacts to last period's deviations from equilibrium and to past changes

$$\Delta y_t = \alpha \beta' y_{t-1} + \sum_{i=1}^{p-1} \Gamma_i \Delta y_{t-i} + \Phi D_t + \varepsilon_t$$

- Notation

- ▶ $y_t \in \mathbb{R}^n$: the n I(1) variables; $\Delta y_t = y_t - y_{t-1}$; p : the lag order of the VAR in levels; T : the sample size
 - ▶ r : the cointegration rank; β ($n \times r$): the cointegrating vectors, $\beta' y_t$ stationary; α ($n \times r$): the adjustment coefficients
 - ▶ Γ_i : short-run matrices; D_t : deterministic terms (constant, trend) with coefficients Φ ; $\varepsilon_t \sim N(0, \Omega)$: innovations
- $H(r)$: the hypothesis that the rank is at most r

Prerequisites for Today (2/8): the rank tests

- Johansen: regress Δy_t and y_{t-1} on the short-run terms; residuals R_{0t}, R_{1t} ; moments $S_{ij} = T^{-1} \sum_t R_{it} R'_{jt}$
 - ▶ eigenvalues $1 > \hat{\lambda}_1 > \dots > \hat{\lambda}_n > 0$ of $S_{11}^{-1} S_{10} S_{00}^{-1} S_{01}$: the squared canonical correlations of Δy_t and y_{t-1} after the short run
 - ▶ a large $\hat{\lambda}_i$: a combination of levels that predicts the changes, i.e. a cointegrating relation
- Trace: $LR_{tr}(r) = -T \sum_{i>r} \ln(1 - \hat{\lambda}_i)$; max-eigenvalue: $LR_{\max}(r) = -T \ln(1 - \hat{\lambda}_{r+1})$
 - ▶ both are large when eigenvalues that $H(r)$ sets to zero are not small; non-standard critical values (simulated)
 - ▶ test $r = 0, 1, \dots$ and stop at the first non-rejection
- Deterministic cases: 1 none; 2 constant restricted to $\beta'y$; 3 unrestricted constant; 4 trend restricted to $\beta'y$; 5 unrestricted trend
 - ▶ each case has its own critical values

Prerequisites for Today (3/8): small samples and LR tests

- Small samples
 - ▶ Reinsel–Ahn: multiply the statistic by $(T - np)/T$, a degrees-of-freedom correction
 - ▶ wild bootstrap: simulate the null from the model estimated under $H(r)$, with residuals times $N(0, 1)$ weights; p-value = share of bootstrap statistics above the observed one
- Identification: $r - 1$ restrictions per vector besides the normalisation; $n_1 =$ rows of β (variables plus restricted terms)
 - ▶ same restriction on all vectors, $\beta = H\varphi$: H ($n_1 \times s$) known, φ ($s \times r$) free; LR degrees of freedom $r(n_1 - s)$
 - ▶ β fully known: $r(n_1 - r)$ degrees of freedom
 - ▶ $\alpha = A\psi$: A ($n \times m$) known, ψ ($m \times r$) free; $r(n - m)$ degrees of freedom
- Weak exogeneity of x_t : the row of α for x_t is zero (r restrictions)
 - ▶ x_t does not adjust to disequilibria, so we may condition on it

Prerequisites for Today (4/8): common trends

- ▣ Granger representation: the levels are a sum of random-walk trends plus stationary terms

$$y_t = C \sum_{s \leq t} \varepsilon_s + \text{stationary terms}, \quad C = \beta_{\perp} (\alpha'_{\perp} \Gamma \beta_{\perp})^{-1} \alpha'_{\perp}$$

- ▶ $\Gamma = I - \sum_i \Gamma_i$; $\alpha_{\perp}$, $\beta_{\perp}$ ($n \times (n - r)$): orthogonal complements, $\alpha'_{\perp} \alpha_{\perp} = 0$, $\beta'_{\perp} \beta_{\perp} = 0$
- ▶ C : the long-run effect of a shock ε_t on the levels; common trends: $\alpha'_{\perp} \sum_s \varepsilon_s$
- ▣ One common trend: the permanent shock and its impact (KPSW: King, Plosser, Stock and Watson)
 - ▶ $u_t^P = \alpha'_{\perp} \varepsilon_t / (\alpha'_{\perp} \Omega \alpha_{\perp})^{1/2}$: the innovation of the common trend, scaled to unit variance
 - ▶ $b^P = \Omega \alpha_{\perp} (\alpha'_{\perp} \Omega \alpha_{\perp})^{-1/2}$: the effect of u_t^P on the n variables on impact

Prerequisites for Today (5/8): ARDL

- ▣ ARDL(p, q): y_t on its own p lags and on x_t with q lags

$$y_t = c + \sum_{i=1}^p \phi_i y_{t-i} + \sum_{j=0}^q \beta'_j x_{t-j} + u_t$$

- ▶ x_t : k regressors; c : intercept; ϕ_i, β_j : coefficients; u_t : error

- ▣ The same model in conditional ECM form

$$\Delta y_t = c + \pi_{yy} y_{t-1} + \pi'_{yx} x_{t-1} + \sum \psi_i \Delta y_{t-i} + \sum \omega'_j \Delta x_{t-j} + u_t$$

- ▶ $\pi_{yy} = \sum_i \phi_i - 1$: the speed of adjustment, negative if y returns to equilibrium; $\pi_{yx} = \sum_j \beta_j$; ψ_i, ω_j : short-run coefficients
- ▶ long run $\theta = -\pi_{yx}/\pi_{yy}$: the equilibrium is $y = \theta'x$ plus a constant
- ▶ half-life of a disequilibrium in an ARDL(1, q): $\ln 0.5 / \ln(1 + \pi_{yy})$ periods

Prerequisites for Today (6/8): the bounds test and NARDL

- ▣ Bounds test (PSS: Pesaran, Shin and Smith 2001): F for $\pi_{yy} = \pi_{yx} = 0$, t for $\pi_{yy} = 0$; H_0 : no level relationship
 - ▶ lower bound: critical value if all x are $I(0)$; upper bound: if all x are $I(1)$
 - ▶ F above the upper bound: a level relationship; below the lower bound: none; between: inconclusive
 - ▶ cases I–V as for Johansen; the bounds depend on the case, on k and, in small samples, on T
- ▣ NARDL (nonlinear ARDL): increases and decreases of x enter separately
 - ▶ $x_t^+ = \sum_{j \leq t} \max(\Delta x_j, 0)$: the cumulated increases; $x_t^- = \sum_{j \leq t} \min(\Delta x_j, 0)$: the cumulated decreases
 - ▶ θ^+ , θ^- : their long-run coefficients; long-run asymmetry: $\theta^+ \neq \theta^-$

Prerequisites for Today (7/8): panel dependence and unit roots

- ▣ **CD** test of cross-section dependence: $CD = \sqrt{2T/(N(N-1))} \sum_{i < j} \hat{\rho}_{ij}$
 - ▶ N units, T periods; $\hat{\rho}_{ij}$: correlation over time of units i and j
 - ▶ $CD \rightarrow N(0, 1)$ under weak dependence; $|CD| > 1.96$ rejects at 5%
- ▣ **IPS** (Im, Pesaran, Shin): $\bar{t} = N^{-1} \sum_i t_i$
 - ▶ t_i : the ADF t -ratio of country i ; assumes independent units
 - ▶ over-rejects under common shocks
- ▣ **CIPS** (Pesaran 2007): each ADF augmented with $\bar{y}_{t-1}$ and $\Delta \bar{y}_t$
 - ▶ $\bar{y}_t = N^{-1} \sum_i y_{it}$: the cross-section average, which removes one common factor
 - ▶ critical values of IPS and CIPS by simulation of independent random walks with the given N and T

Prerequisites for Today (8/8): heterogeneous panel estimators

- ▣ **MG** (mean group): $\hat{\theta}_{MG} = N^{-1} \sum_i \hat{\theta}_i$
 - ▶ $\hat{\theta}_i$: the long-run estimate of country i
 - ▶ $SE = s/\sqrt{N}$; s : the standard deviation of the $\hat{\theta}_i$
- ▣ **PMG** (pooled mean group)
 - ▶ common long run, country-specific short run; efficient if the long run is homogeneous
- ▣ Hausman test: $H = (\hat{\theta}_{MG} - \hat{\theta}_{PMG})'(V_{MG} - V_{PMG})^{-1}(\hat{\theta}_{MG} - \hat{\theta}_{PMG})$
 - ▶ V_{MG} , V_{PMG} : the estimated covariance matrices; k : the number of long-run coefficients
 - ▶ $H \rightarrow \chi^2(k)$ under long-run homogeneity; a large H rejects the common long run

A1: the trace test from eigenvalues [Solved]

Task

- ▣ $n = 3$, VAR(2) in levels, case 3, effective $T = 100$; eigenvalues $\hat{\lambda} = (0.30; 0.12; 0.03)$; 95% quantiles of the trace statistic for $n - r = 3, 2, 1$: 29.78, 15.48, 3.89.
- ▣ 1. Compute the trace statistics for $r = 0, 1, 2$ and the max-eigenvalue statistics.
- ▣ 2. Determine the rank by the sequential procedure.
- ▣ 3. Apply the Reinsel–Ahn factor and say whether the rank changes.
- ▣ Report: six statistics, one rank, one sentence.

Solution

- ▣ 1. $-100 \ln(0.70) = 35.67$, $-100 \ln(0.88) = 12.78$, $-100 \ln(0.97) = 3.05$; trace: 51.50, 15.83, 3.05; max-eigenvalue: 35.67, 12.78, 3.05
- ▣ 2. $51.50 > 29.78$: reject $r = 0$; $15.83 > 15.48$: reject $r \leq 1$ (narrowly); $3.05 < 3.89$: $\hat{r} = 2$
- ▣ 3. Factor $(100 - 6)/100 = 0.94$: 48.41, 14.88, 2.86; now $14.88 < 15.48$, so $\hat{r} = 1$: the second relation does not survive the correction

A2: degrees of freedom of LR tests [Proposed]

Solution discussed in class

Task

- ▣ $n = 4$ variables, case 3 ($n_1 = 4$), rank $r = 2$. Three hypotheses with their LR statistics. Model: A1.
- ▣ 1. Variable 4 is excluded from both relations, $\beta = H\varphi$ with H of dimension 4×3 : give the degrees of freedom and decide at 5% for $LR = 3.1$.
- ▣ 2. Variable 1 is weakly exogenous: give the degrees of freedom and decide for $LR = 7.4$.
- ▣ 3. Both vectors are fully known from theory: give the degrees of freedom and decide for $LR = 9.0$.
- ▣ 4. How many restrictions just-identify β , and why do they not change the likelihood?
- ▣ Report: three degrees of freedom with decisions, one number, one sentence.

A3: the Granger representation by hand [Solved]

Task

- ▣ Bivariate VECM without lags: $\Delta y_t = \alpha \beta' y_{t-1} + \varepsilon_t$, $\alpha = (-0.2, 0.1)'$, $\beta = (1, -1)'$, so $\Gamma = I$.
- ▣ 1. Find $\alpha_\perp$ and $\beta_\perp$ (first element normalised to 1).
- ▣ 2. Compute $C = \beta_\perp (\alpha'_\perp \beta_\perp)^{-1} \alpha'_\perp$ and check $\beta' C = 0$ and $C \alpha = 0$.
- ▣ 3. Write the common trend and explain which variable carries more of it.
- ▣ Report: two vectors, one matrix, two checks, one sentence.

Solution

- ▣ 1. $\alpha'_\perp \alpha_\perp = 0$: $-0.2a + 0.1b = 0$, $\alpha_\perp = (1, 2)'$; $\beta_\perp = (1, 1)'$
- ▣ 2. $\alpha'_\perp \beta_\perp = 3$, $C = \frac{1}{3} \begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix}$; $\beta' C = (0, 0)$,
 $C \alpha = \frac{1}{3}(-0.2 + 0.2)(1, 1)' = 0$
- ▣ 3. Trend $\frac{1}{3} \sum_s (\varepsilon_{1s} + 2\varepsilon_{2s})$, common to both: y_2 adjusts less ($|\alpha_2| < |\alpha_1|$), so its shocks weigh twice as much in the trend

A4: the permanent shock of a common-trends model [Proposed]

Solution discussed in class

Task

- ▣ $n = 3$, $r = 2$, $\alpha = \begin{pmatrix} -0.1 & 0 \\ 0 & -0.2 \\ 0.05 & 0.1 \end{pmatrix}$, $\Omega = \begin{pmatrix} 1 & 0.3 & 0.5 \\ 0.3 & 4 & 1 \\ 0.5 & 1 & 1 \end{pmatrix}$. Model: A3.
- ▣ 1. Find $\alpha_{\perp}$ (first element 1).
- ▣ 2. Compute $\alpha'_{\perp} \Omega \alpha_{\perp}$ and the impact vector b^P of the permanent shock.
- ▣ 3. Compute the share of the permanent shock in the one-step forecast error variance of each variable, $(b_i^P)^2 / \Omega_{ii}$, and interpret.
- ▣ Report: one vector, one number, one vector, three shares.

A5: an ARDL in error-correction form [Solved]

Task

- ▣ ARDL(1, 1): $y_t = 0.5 + 0.7y_{t-1} + 0.2x_t + 0.1x_{t-1} + u_t$.
- ▣ 1. Rewrite it as a conditional ECM; give π_{yy} , π_{yx} and the long-run coefficient θ .
- ▣ 2. Compute the half-life of a disequilibrium.
- ▣ 3. Compute the cumulative multipliers of a permanent unit rise of x after 0, 1, 2 and 3 periods.
- ▣ Report: one equation, three numbers, one number, four numbers.

Solution

- ▣ 1. $\Delta y_t = 0.5 - 0.3y_{t-1} + 0.3x_{t-1} + 0.2\Delta x_t + u_t = 0.5 - 0.3(y_{t-1} - x_{t-1}) + 0.2\Delta x_t + u_t$; $\theta = 0.3/0.3 = 1$, long-run intercept $0.5/0.3 = 1.667$
- ▣ 2. $\ln 0.5 / \ln 0.7 = 1.94$ periods
- ▣ 3. $m_0 = 0.2$; $m_h = 0.7m_{h-1} + 0.3$: 0.200, 0.440, 0.608, 0.726 $\rightarrow \theta = 1$

A6: a bounds-test decision [Proposed]

Solution discussed in class

Task

- ▣ Case III, $k = 1$, $T = 60$ annual observations. 5% bounds: asymptotic $[4.97; 5.71]$, simulated for $T = 60$ $[5.18; 5.96]$; t -bounds for $T = 60$ $[-2.87; -3.26]$. Model: A5.
- ▣ 1. For three studies with $F = 5.4$, $F = 5.8$ and $F = 6.1$, give the decision with the asymptotic and with the small-sample bounds.
- ▣ 2. In the study with $F = 6.1$, $t = -2.9$. What do you conclude?
- ▣ 3. Which study changes its conclusion when the correct bounds are used?
- ▣ Report: six decisions, one sentence, one study.

A7: the CD statistic by hand [Solved]

Task

- ▣ $N = 4$ countries, $T = 25$; pairwise residual correlations $\hat{\rho}_{12} = 0.6$, $\hat{\rho}_{13} = 0.5$, $\hat{\rho}_{14} = 0.4$, $\hat{\rho}_{23} = 0.7$, $\hat{\rho}_{24} = 0.3$, $\hat{\rho}_{34} = 0.5$.
- ▣ 1. Compute the CD statistic and its decision at 5%.
- ▣ 2. Explain what a CD statistic close to zero would and would not prove.
- ▣ 3. Say which panel unit-root test is appropriate after this result.
- ▣ Report: one statistic, two sentences, one test.

Solution

- ▣ 1. $\sum \hat{\rho}_{ij} = 3.0$; $CD = \sqrt{2 \cdot 25 / (4 \cdot 3)} \cdot 3.0 = 6.12 > 1.96$: reject weak dependence
- ▣ 2. Near zero: no average correlation; it does not prove independence, because positive and negative $\hat{\rho}_{ij}$ can cancel (report the mean $|\hat{\rho}_{ij}|$ too)
- ▣ 3. CIPS (cross-sectionally augmented), not LLC or IPS

A8: the Hausman test of PMG against MG [Proposed]

Solution discussed in class

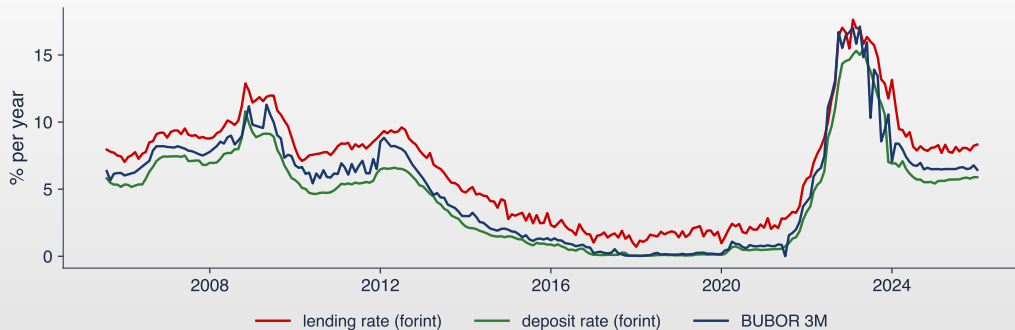
Task

- ▣ Long-run income elasticity: MG 0.85 (SE 0.10), PMG 0.95 (SE 0.04). Four country estimates: 0.6, 0.9, 1.0, 0.9. Model: A7.
- ▣ 1. Compute the Hausman statistic and its p -value.
- ▣ 2. Compute the MG estimate and its standard error from the four country estimates.
- ▣ 3. Explain why $V_{MG} - V_{PMG}$ must be positive and what you would do if it were not.
- ▣ Report: two numbers, two numbers, two sentences.

B1: the rank of the Hungarian pass-through system [Solved]

- **Question:** do Hungarian bank lending and deposit rates share long-run relations with the money-market rate?
- **Inputs:** lending and deposit rates in forint (IMF IFS), BUBOR 3M (Eurostat; isolated missing months interpolated), monthly, $T = 246$; case 2
- **Tasks**
 1. Select the lag order of the VAR in levels by AIC, BIC and HQ (up to 8).
 2. Compute the trace statistics with the BIC lag order and compare them with the simulated 95% quantiles of case 2.
 3. Apply the Reinsel–Ahn correction and compute wild-bootstrap p -values with 399 samples.
 4. Interpretation: is the rank robust to the small-sample corrections?
- **Report:** three lag orders, a table of three rows, one rank, one sentence
- **Notebook:** section B1

B1: solution (1/2) [Solved]



B1: solution (2/2) [Solved]

$H(r)$	LR_{tr}	RA	cv 5%	wild p
$r = 0$	81.4	77.4	35.2	0.003
$r \leq 1$	31.2	29.6	20.5	0.005
$r \leq 2$	3.9	3.7	9.3	0.390

- Lags: AIC 8, BIC 4, HQ 4; $p = 4$; eigenvalues 0.188, 0.106, 0.016; all three methods give $\hat{r} = 2$
- Interpretation: the Hungarian rank survives both small-sample corrections
 - ▶ two cointegrating relations and one common trend
 - ▶ with a shorter sample or more lags the corrections can overturn the asymptotic rank: report all three

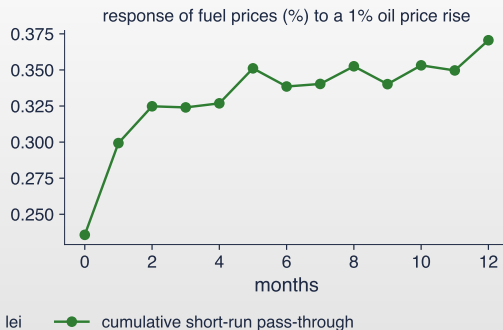
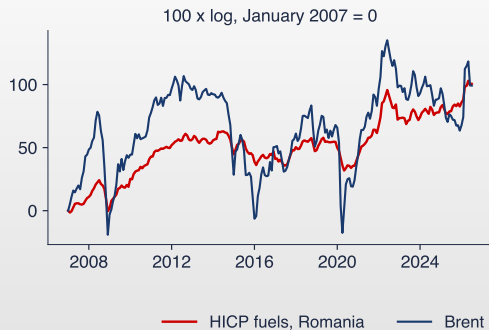
B2: tests on β and α for Hungary [Proposed]

- **Question:** is the long-run pass-through to Hungarian lending and deposit rates complete, and does BUBOR adjust to bank rates?
- **Inputs:** the data and the VECM of B1 with $r = 2$, case 2; model: B1
- **Tasks**
 1. Just-identify β : the lending relation excludes the deposit rate, the deposit relation excludes the lending rate; report both relations.
 2. Test complete pass-through to both rates (coefficient -1 on BUBOR in both vectors), then to the lending rate alone; give the degrees of freedom.
 3. Test weak exogeneity of BUBOR (a zero row of α) and report $\hat{\alpha}$.
 4. Interpretation: can a single-equation ARDL for the lending rate be used here?
- **Report:** two relations, three LR tests with degrees of freedom and p -values, one matrix, one sentence
- **Notebook:** section B2

B3: oil prices and Romanian fuel prices, bounds test [Solved]

- **Question:** is there a long-run relation between Romanian retail fuel prices and the oil price in lei?
- **Inputs:** y : 100 ln HICP fuels and lubricants (CP0722, Eurostat); x : 100 ln (Brent in USD, FRED $\times$ USD/RON, BNR monthly average); monthly, $T = 235$
- **Tasks**
 1. Select an ARDL by BIC (up to 6 lags each) in case III and estimate the conditional ECM.
 2. Compute the bounds F and t and compare them with asymptotic and small-sample 5% bounds.
 3. Repeat in case V (trend) and report the long-run elasticity in both cases.
 4. Interpretation: what does the absence of a level relation mean for fuel prices?
- **Report:** two models, two F and two t with bounds, two elasticities, one sentence
- **Notebook:** section B3

B3: solution (1/2) [Solved]



B3: solution (2/2) [Solved]

- Case III, ARDL(1, 2): $F = 1.79$ against $[4.97; 5.71]$ (asymptotic) and $[5.06; 5.86]$ (T of the sample); $t = -1.27$ against $[-2.86; -3.22]$; $\hat{\theta} = 1.07$ (SE 0.54)
- Case V: $F = 4.05$ against $[6.52; 7.41]$; $t = -2.49$ against $[-3.41; -3.69]$; $\hat{\theta} = 0.51$ (SE 0.10)
- Short run (right panel): a 1% rise in the oil price raises fuel prices by 0.24% in the same month and by 0.37% after 12 months
- Interpretation: F is below the lower bound in both cases, so there is no stable level relation
 - ▶ excise duties, VAT changes and the 2022 price caps shift the retail price independently of oil
 - ▶ $\hat{\theta}$ is therefore not interpretable as a long-run elasticity

B4: the KPSW common trend on longer samples [Proposed]

- **Question:** does the share of the permanent (balanced-growth) shock in US output fluctuations survive a longer sample and a trend in the great ratios?
- **Inputs:** 100 ln real per capita consumption of nondurables and services, fixed investment and GDP (FRED), quarterly, from 1949Q1; model: B1 and A4
- **Tasks**
 1. For samples ending in 1988Q4, 2019Q4 and 2025Q4, test the great ratios $\beta = ((1, 0, -1)', (0, 1, -1)')$ in case 3 and, with a restricted trend, in case 4.
 2. Impose the great ratios and compute the share of the permanent shock in the forecast error variance of output at 1, 8 and 24 quarters, in both cases.
 3. Interpretation: is the importance of trend shocks a property of the data or of the specification?
- **Report:** a table of six rows (sample and case) with an LR test and three shares, one sentence
- **Notebook:** section B4

B5: EU-27 inflation, IPS against CIPS [Proposed]

- **Question:** is annual inflation stationary in the EU-27 once the common shocks are taken into account?
- **Inputs:** annual HICP inflation, Eurostat `prc_hicp_aind`, $N = 27$, 1999–2025 (balanced); model: A7 and B1
- **Tasks**
 1. Compute the CD statistic of the demeaned series and the mean absolute correlation.
 2. Compute IPS $\bar{t}$ and CIPS with one lag and a constant; obtain 5% critical values and p -values from 1,000 simulations of independent random walks with the same N and T .
 3. Repeat from 2000 onward.
 4. Interpretation: does cross-section dependence change the conclusion here?
- **Report:** a table of two rows with CD, mean $|\rho|$, IPS and CIPS with p -values, one sentence
- **Notebook:** section B5

C1: rockets and feathers in Romanian fuel prices [Proposed]

- ▣ **Question:** do Romanian fuel prices react more to oil-price increases than to decreases?
- ▣ **Inputs:** the data of B3; partial sums x_t^+ , x_t^- of the changes of the oil price in lei; model: B3
- ▣ **Tasks**
 1. Write a five-line pre-registration: hypothesis, variables, case, lag rule, the two asymmetry tests.
 2. Estimate the NARDL in case III with lags by BIC (up to 4); report the bounds F with $k = 2$ and the long-run coefficients θ^+ , θ^- .
 3. Test long-run symmetry and impact symmetry by Wald tests.
 4. Interpretation: what can and cannot be concluded, and how would you turn this into a project?
- ▣ **Report:** a five-line plan, one F , two coefficients, two Wald tests, a project plan
- ▣ **Notebook:** section C1

C2: audit an AI answer [Proposed]

- ▣ A student asked an AI assistant to summarise cointegration testing. The answer:
- ▣ (a) For two interest rates without drift, use the case-3 critical value 3.84 for $H(1)$.
- ▣ (b) The eigenvalues of the Johansen problem are squared canonical correlations.
- ▣ (c) With $n = 4$ and $r = 2$, testing two fully known cointegrating vectors has 2 degrees of freedom.
- ▣ (d) ARDL bounds tests are valid whatever the integration order of the regressors, including $I(2)$.
- ▣ (e) A weakly exogenous variable has a zero row in α , so it can be conditioned on.
- ▣ (f) If the IPS test rejects a panel unit root, every country is stationary.
- ▣ (g) The bootstrap rank test should resample from the unrestricted VAR in levels.
- ▣ Tasks
 - ▶ 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number (notebook section C2).
 - ▶ 2. Report: a list of seven verdicts with one line of justification each.

Key takeaways

- ▣ The rank comes from eigenvalues, but its critical values depend on the deterministic case and on the sample size
- ▣ Count the degrees of freedom of every test on β and α by hand
- ▣ An ARDL needs weak exogeneity and correct bounds; the absence of a level relation is a result too
- ▣ In panels, measure the cross-section dependence before trusting a unit-root or cointegration test
- ▣ An AI answer is a draft: check the case, the degrees of freedom, the alternative hypothesis and the bootstrap design

After the seminar

- ▣ Lecture 4 develops each topic of today: the likelihood behind Johansen, the five cases, the bootstrap, identification, common trends, ARDL, panels
- ▣ Try the [Proposed] tasks in the notebook
- ▣ C1 can grow into a team project: asymmetric price transmission with weekly data
- ▣ Reading: [Johansen \(1991\)](#); [Pesaran, Shin and Smith \(2001\)](#); [Pesaran, Shin and Smith \(1999\)](#); [Pesaran \(2007\)](#)
- ▣ **The seminar is for practice and is not graded; the solutions of [Proposed] tasks are discussed in class**

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